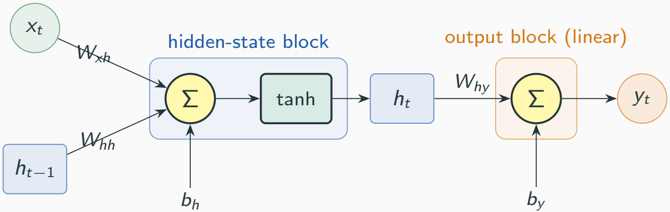
Recurrent Neural Network (RNN)

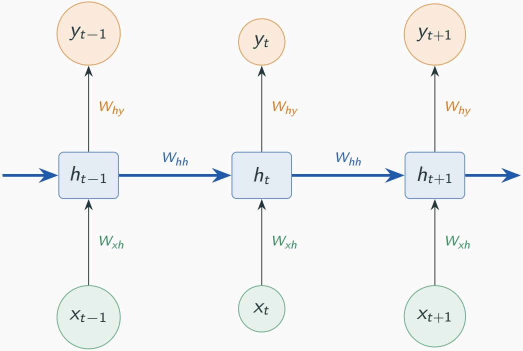


$$h_t = \tanh\big(W_{hh} h_{t-1} + W_{xh} x_t + b_h\big)$$

$$y_t = W_{hy} h_t + b_y$$

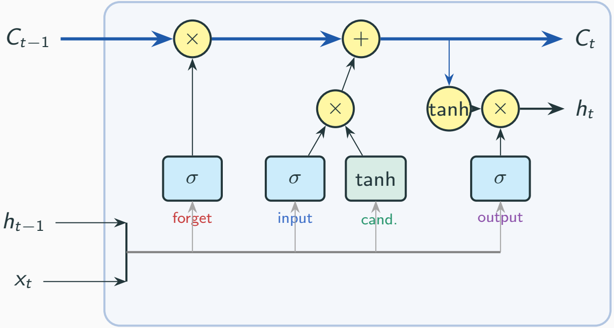
Σ is a weighted sum of its inputs plus a bias. Only the hidden block applies a tanh; the output block is purely linear. The weights W_{xh}, W_{hh}, W_{hy} are shared across every time step.

RNN - Unrolled Across Time



- The same cell is applied at every step; h_t carries information forward as memory.
- Weights W_{xh}, W_{hh}, W_{hy} are shared across all time steps.
- Trained by back-propagation through time (BPTT).
- Long dependencies $\Rightarrow$ vanishing / exploding gradients (motivating LSTM & GRU).

LSTM - Cell Architecture



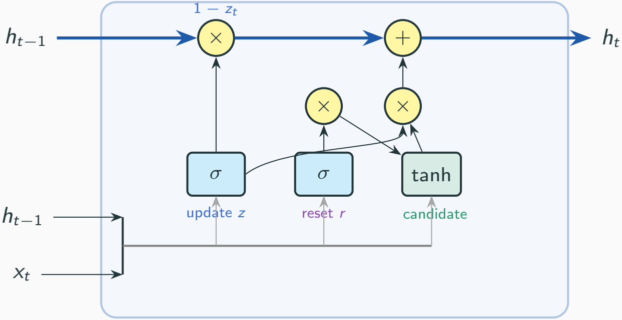
The horizontal cell-state highway $C_{t-1} \rightarrow C_t$ carries long-term memory; three **gates** (σ) regulate what to forget, write, and output, while a tanh forms the candidate.

LSTM - Equations

$$
\begin{aligned}
f_t &= \sigma\big(W_f[h_{t-1}, x_t] + b_f\big) && \text{forget gate} \\
i_t &= \sigma\big(W_i[h_{t-1}, x_t] + b_i\big) && \text{input gate} \\
\tilde{C}_t &= \tanh\big(W_C[h_{t-1}, x_t] + b_C\big) && \text{candidate} \\
C_t &= f_t \odot C_{t-1} + i_t \odot \tilde{C}_t && \text{cell-state update} \\
o_t &= \sigma\big(W_o[h_{t-1}, x_t] + b_o\big) && \text{output gate} \\
h_t &= o_t \odot \tanh(C_t) && \text{hidden state} \\
y_t &= W_{hy}\, h_t + b_y && \text{output}
\end{aligned}
$$

- $\sigma \in (0, 1)$ gates the flow; $\odot$ is element-wise multiplication.
- The additive cell update lets gradients flow far back, easing vanishing gradients.

GRU - Cell Architecture



Only *one* state h (no separate cell). The update gate z interpolates between the old state and a new candidate; the reset gate r controls how much past to use when forming it.

GRU - Equations

$$
\begin{aligned}
z_t &= \sigma\big(W_z[h_{t-1}, x_t] + b_z\big) && \text{update gate} \\
r_t &= \sigma\big(W_r[h_{t-1}, x_t] + b_r\big) && \text{reset gate} \\
\tilde{h}_t &= \tanh\big(W[\,r_t \odot h_{t-1},\ x_t\,] + b\big) && \text{candidate} \\
h_t &= (1 - z_t) \odot h_{t-1} + z_t \odot \tilde{h}_t && \text{new state} \\
y_t &= W_{hy}\, h_t + b_y && \text{output}
\end{aligned}
$$

- Merges cell & hidden state $\Rightarrow$ fewer parameters than LSTM (3 weight matrices vs. 4).
- $z_t \to 0$ keeps the old state; $z_t \to 1$ overwrites it - the two terms form a *convex combination*.
- Often matches LSTM accuracy at lower computational cost.